Andrew McLoughlin

Summary

Full-stack developer of scientific software with an academic foundation in computational biology and epidemiology (MPhil from Cambridge, MSc/BSc from Warwick). Over a decade of experience across healthcare-related start-ups specializing in building robust data pipelines, analytics applications, clinical data systems, and public-facing scientific databases (such as the Gates Foundation-funded VOC Atlas). Combines technical production-grade expertise in Python, Django, PostgreSQL, and dbt with deep knowledge of clinical data management, compliance (GDPR, GCP, 21 CFR Part 11, CDISC/CDASH), and product ownership.

Experience

Senior Data Engineer & Product Owner

Owlstone Medical
2020-08 – present

- Product Owner and lead developer for www.vocatlas.com, a greenfield public-facing Django application helping scientists identify compounds on breath across populations, funded in part by the Bill & Melinda Gates Foundation.
- Managed cross-functional teams (scientists, designers, developers, QC, marketing), maintaining Jira backlogs, tracking OKRs, defining user personas, and leading stakeholder meetings with the Gates Foundation.
- Designed and deployed a lightweight internal order fulfillment management application with interactive analytics dashboards integrated with Royal Mail shipping API, reducing data entry error rates from 5% to negligible levels and completely obsolescing a 2-hour weekly meeting.
- Set up the company Data Team to tackle inefficient/non-compliant data practices and championed the adoption of a single company-wide open-source Data Warehouse (using Airbyte, Airflow, dbt, PostgreSQL, and SQL) following Kimball's toolkit best practices.
- Built and automated ETL pipelines ingesting millions of rows of GCMS data, connecting third-party APIs, and feeding interactive business intelligence dashboards used by hundreds of employees to drive key leadership decisions.
- Led UX/UI design of the OMED mobile app and website, building interactive prototypes in Figma and Penpot to quickly iterate on user feedback.
- Maintained a monolithic Django-based Laboratory Management Information System (LIMS) used daily by over 200 employees with 100% uptime over 8 years, utilizing Kubernetes, Docker/containers, PostgreSQL, Gitlab CI/CD, Pytest, and Sentry.
- Ensured rigorous data governance, security, and compliance with GDPR, GCP, and 21 CFR Part 11 via automated immutable audit logging, SSO, and MFA minimal access controls.
- Productionised statisticians' research code into tested, version-controlled Python packages with strict automated testing and 100% code coverage.

Lead Clinical Data Manager

Owlstone Medical
2017-12 – 2020-08

- Managed clinical trial data (such as the LuCID clinical trial, a large-scale prospective case-control study evaluating breath-based biomarkers for lung cancer over a 24-month follow-up across 26 clinical sites in 5 countries).
- Applied Multivariate Pattern Recognition and Classification Algorithms in Python to differentiate signals between cases and controls, presenting insights via PowerBI.
- Designed CDISC/CDASH-compatible Case Report Forms (CRFs) using SNOMED-CT and MedDRA clinical terminologies, preparing datasets for FDA submission and cross-study aggregation.

Clinical Data Manager / Data Manager

Optimum Patient Care (Research in Real Life)
2013-08 – 2017-12

- Designed and built CRFs in OpenClinica, enforcing robust data quality with edit checks, skip logic, and comprehensive documentation.
- Built interactive dashboards to report study recruitment metrics and endpoints to study managers, enabling early identification and resolution of demographic imbalances across study arms.
- Ingested and processed complex real-world data into OPCRD, executing data validation and cleaning scripts.
- Transitioned clinical teams from paper-based operations to fast, efficient, and compliant digital systems across the EU, UK, and US for over 10 clinical trials.

Skills

Languages & Core

- Python 3.12, SQL, R, JavaScript, VBA

Backend & Web Frameworks

- Django, Pydantic, APIs, Pytest, Bootstrap, Sentry, Allauth

Data Engineering & Warehousing

- dbt, Airflow, Airbyte, PostgreSQL, MS SQL Server, Google BigQuery, Rundeck, ETL/ELT Architecture, Kimball Data Modeling

Data Science & Statistics

- Pandas, NumPy, Statistical Modelling, Multivariate Pattern Recognition, Signal Processing, Clinical Data Analysis

DevOps, Infra & Version Control

- Kubernetes, Docker, Git, GitLab CI/CD, Linux

Clinical Data Management & Standards

- OpenClinica, RedCap, CDISC, CDASH, SNOMED-CT, MedDRA, OPCRD, ISAR, CRF Design, GDPR, GCP, 21 CFR Part 11

BI & Data Visualization

- Metabase, Google Looker Studio, PowerBI, Tableau / BI Dashboards

Product & Design

- Product Ownership, Agile / Scrum (Certified Scrum Master), Jira, Figma, Penpot, UX/UI Prototyping, User Interviewing & Shadowing

Education

MPhil in Statistical Epidemiology / Cardiovascular Epidemiology

University of Cambridge
MRC / Pfizer Scholarship
2013 – 2013

MSc in Computational Biology / Computational Systems Biology

University of Warwick
Systems Biology Studentship
2012 – 2012

BSc in Biomedical Science

University of Warwick
2.1 Hons
2011 – 2011

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